```c
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int source, destination;
    char buffer[100];
    int bytesRead;

    source = open("sample.txt", O_RDONLY);

    if (source == -1)
    {
        printf("Error opening source file.\n");
        return 1;
    }

    destination = open("copy.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (destination == -1)
    {
        printf("Error creating destination file.\n");
        close(source);
        return 1;
    }
```

```
himasree@LAPTOP-41COT5AA:~/OSLab$ nano sample.txt
himasree@LAPTOP-41COT5AA:~/OSLab$ gcc q2.c -o q2
himasree@LAPTOP-41COT5AA:~/OSLab$ ./q2
File copied successfully.
himasree@LAPTOP-41COT5AA:~/OSLab$ cat copy.txt
Hello OSSP
Operating Systems Lab
himasree@LAPTOP-41COT5AA:~/OSLab$ strace cat sample.txt
Command 'strace' not found, but can be installed with:
sudo apt install strace
himasree@LAPTOP-41COT5AA:~/OSLab$ sudo apt update
sudo apt install strace
[sudo] password for himasree:
Sorry, try again.
[sudo] password for himasree:
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [897 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1174 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [198 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [46.4 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1199 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [280 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [181 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1681 kB]
Get:13 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [239 kB]
Get:14 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [76.3 kB]
```

```
himasree@LAPTOP-41COT5AA:~/OSLab$ strace cat sample.txt
execve("/usr/bin/cat", ["cat", "sample.txt"], 0x7fff2bee0438 /* 26 vars */) = 0
brk(NULL)                               = 0x5803799f2000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x747efe66d000
access("/etc/ld.so.preload", R_OK)      = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=25767, ...}) = 0
mmap(NULL, 25767, PROT_READ, MAP_PRIVATE, 3, 0) = 0x747efe666000
close(3)                                = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\220\243\2\0\0\0\0"..., 832) = 832
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..., 784, 64) = 784
fstat(3, {st_mode=S_IFREG|0755, st_size=2125328, ...}) = 0
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..., 784, 64) = 784
mmap(NULL, 2170256, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x747efe400000
mmap(0x747efe428000, 1605632, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x747efe428000
mmap(0x747efe5b0000, 323584, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1b0000) = 0x747efe5b0000
mmap(0x747efe5ff000, 24576, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1fe000) = 0x747efe5ff000
mmap(0x747efe605000, 52624, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x747efe605000
close(3)                                = 0
mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x747efe663000
```